2.1 Warm-up: Mental Math Solutions

Students write only the answer. The notes give the step a student should be doing in their head, and the mistakes worth looking for.

1. Fill in the blanks.

   (a) 2.5 km = **2500 m**

   (b) 450 cm = **4.5 m**

   (c) 0.8 m = **80 cm**

   (d) 35 mm = **3.5 cm**

   (e) 1 m = **1000 mm**

   (f) 7 mm = **0.007 m**

   Going to a smaller unit makes the number bigger. A student who writes 0.0025 m for (a) or 45,000 m for (b) has moved the decimal point the wrong way.

2. **10,000,000 m** (ten million meters).
   So the distance around the Earth through the poles is about 40,000 km. The modern value is 40,008 km — the 1790s survey was very good.

3. **100 sheets.** 1 cm = 10 mm, and $10 \div 0.1 = 100$.
   A 500-sheet ream of paper is about 5 cm thick, which students can check.

4. Phone storage.
   (a) **256,000 MB.** 1 GB = 1000 MB.

   (b) **64,000 photos.** $256{,}000 \div 4 = 64{,}000$.
   Phone and drive makers use 1 GB = 1000 MB. Some software uses 1024, so a student who knows that may write 262,144 MB; accept it, but it is not mental math.

5. **4,000,000,000** ticks per second (four billion).
   Giga means a billion, nine zeros. One hertz is one cycle per second.

6. Micrometers.
   (a) **1000 µm** in 1 mm. 1 m = 1,000,000 µm and 1 m = 1000 mm.

   (b) **125 cells.** $1000 \div 8 = 125$.

   (c) **10,000 µm.** 1 cm = 10 mm, and $10 \times 1000 = 10{,}000$.
   About 1250 red blood cells would fit across the chip. The transistors on it are smaller again — hundreds of them would fit across one red blood cell.

7. Fractions as decimals.

   (a) $\frac{1}{4} = \mathbf{0.25}$ (c) $\frac{1}{5} = \mathbf{0.2}$ (e) $\frac{1}{8} = \mathbf{0.125}$ (g) $\frac{1}{3} = \mathbf{0.333\ldots}$

   (b) $\frac{3}{4} = \mathbf{0.75}$ (d) $\frac{1}{10} = \mathbf{0.1}$ (f) $\frac{5}{8} = \mathbf{0.625}$ (h) $\frac{2}{3} = \mathbf{0.666\ldots}$

   Accept $0.\overline{3}$ *and* $0.\overline{6}$*, and accept 0.33 or 0.67 written with* $\approx$*. Mark "*$\frac{2}{3} = 0.66$*" wrong: it is neither exact nor correctly rounded.* $\frac{1}{8}$ *is half of* $\frac{1}{4}$*, and* $\frac{5}{8} = 5 \times 0.125$*.*

8. Fractions of an hour and a day.

   (a) $\frac{1}{4}$ hour = **15 min**

   (b) $\frac{1}{3}$ hour = **20 min**

   (c) 0.1 hour = **6 min**

   (d) 30 s = **0.5 min**

   (e) 45 min = **0.75 h**

   (f) 6 hours = $\mathbf{\frac{1}{4}}$ **day**

   (g) 8 hours = $\mathbf{\frac{1}{3}}$ **day**

   (h) 1.2 hours = **1 h 12 min**

   Part (h) is the one to watch: 1.2 h is not 1 h 20 min, because $0.2 \times 60 = 12$*. A clock is not a decimal.*

9. **3600 s** in an hour **365 days** in a year **52 weeks** in a year

   $60 \times 60 = 3600$*. A year is* $365\frac{1}{4}$ *days on average, which is why we have leap years; accept 365 or 366.* $365 \div 7 = 52\frac{1}{7}$*, so "about 52."*

10. Sunlight.

    (a) **500 s.** $150{,}000{,}000 \div 300{,}000 = 1500 \div 3 = 500$.

    (b) **8 min 20 s.** $8 \times 60 = 480$, and $500 - 480 = 20$.

    Cancel the five zeros the two numbers share. In 1-9CW the precise values gave 8.31 min; these round values give $8\frac{1}{3}$ *min.*

11. Circle the best answer.

    (a) **(B) 2.54.** *Exact, by definition. The distractors are other conversion factors: 1.6 km in a mile, 3.28 ft in a meter, 10 mm in a cm.*

    (b) **(C)** $\mathbf{\frac{5}{8}}$**.** *1 km* $\approx$ *0.621 mi, and* $\frac{5}{8} = 0.625$*. Choice (D) is backwards: a mile is about 1.6 km, so a kilometer is less than a mile.*

12. **About 3 miles.** $5 \times \frac{5}{8} = \frac{25}{8} = 3\frac{1}{8}$.

    Accept 3 or 3.1 mi; the true value is 3.11 mi.